

Frobenius-Norm Orthogonal-Projection Variations on CENSOR

A Mathematical, Machine-Learning, and Privacy Analysis of Gradient-Inversion Defenses, with 32 Empirically Evaluated Variants

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Reference paper: Zhang, Cheng, Shen, Ribeiro, An, Chen, Zhang, Li. “CENSOR: Defense Against Gradient Inversion via Orthogonal Subspace Bayesian Sampling.” NDSS 2025.
kaiyuanzhang.com/publications/NDSS25_Censor.pdf

Companion artifacts: `sensor_frobenius_variations.py`,
`sensor_frobenius_variations.ipynb`, `sensor_kaggle_standalone.ipynb`

Abstract

Federated learning avoids sending raw client data to a server, but the *gradients* clients do send can still be inverted to recover private images and, more reliably, private labels. CENSOR (NDSS 2025) defends against this by replacing a client’s true gradient with a randomly-sampled update that is mathematically guaranteed to be exactly orthogonal to it, rescaled to match its original layer-wise magnitude, and chosen from several trial candidates by a cheap local loss check — a construction the authors frame as sampling from a “cold” Bayesian posterior. This report does three things. First, it explains CENSOR’s mechanism, threat model, and published evaluation in full — including the parts of the paper’s evaluation this project does *not* attempt to reproduce (five real image-reconstruction attacks scored by MSE/LPIPS/PSNR/SSIM on ResNet-18) — so the scope of everything that follows is unambiguous. Second, it documents 32 defense variants built around CENSOR’s core orthogonal-projection primitive, organized into four mathematical design axes (the distribution of the injected random candidate, the statistic that gets normalized, the trial-selection rule, and structural/hybrid combinations with other published defenses), each evaluated against a fast label-inference proxy attack on CIFAR-10. Third, it reports the actual measured results from that evaluation — including two negative controls that leak 100% of labels by construction, an accidental block-alignment leak discovered empirically, and a first-order mathematical argument (derived while auditing this project’s own trials-ablation data) explaining why CENSOR’s loss-based trial selection is fundamentally a *second-order* signal at small update learning rates. We conclude that none of the 100% numbers in this study represent a break of CENSOR’s real, paper-reported protection; they either restate the pre-CENSOR vulnerability or come from variants deliberately constructed to demonstrate what a defense must *not* do.

1. Introduction and Scope

1.1 What problem is being studied

In cross-device federated learning (FL), a client never sends its raw training data to the server — only a gradient computed on that data. It has been known since Zhu et al.’s *Deep Leakage from Gradients* (2019) that this gradient alone can be enough to reconstruct the client’s private input, and since Zhao et al.’s *iDLG* (2020) that the private *label* in particular can often be recovered exactly and cheaply, with no optimization required at all. CENSOR (Zhang et al., NDSS 2025) is a recent, carefully evaluated defense against this class of attacks, and it is the subject of this study.

1.2 What this report is, precisely

This report is in three parts:

1. **A faithful explanation of CENSOR itself** (Sections 2–3): the threat model, the two empirical observations that motivate its design, the exact three-phase mechanism (with its underlying equations reproduced and explained), and the paper’s own evaluation methodology and headline results against five real attacks and four baseline defenses.
2. **An extended empirical study built around CENSOR’s core primitive** (Sections 4–6): 32 defense variants, organized into four mathematical design axes, each explained mathematically and evaluated on CIFAR-10 using a fast, narrow label-inference proxy attack rather than the paper’s expensive image-reconstruction attacks.
3. **A full analysis of the actual results obtained**, including several counterintuitive findings that emerged only from close inspection of the data (Section 7), a discussion of what these results do and do not say from a mathematical, machine-learning, and privacy standpoint (Section 8), stated limitations (Section 9), future directions (Section 10), and a summary (Section 11).

1.3 What this report is not

This is not a reproduction of the CENSOR paper’s Table I. The paper evaluates five real gradient-inversion attacks (two stochastic-optimization attacks, IG and GI, and three GAN-based attacks, GGL, GIAS, and GIFD) on a randomly-initialized ResNet-18 across ImageNet, FFHQ, and CIFAR-10, scoring the *reconstructed images* with MSE, LPIPS, PSNR, and SSIM. Running even one of those attacks requires either an iterative optimization loop over pixel space or a pre-trained BigGAN/StyleGAN2 generator, and running all five against 32 defense variants was not feasible for this project. Instead, this study uses a much cheaper diagnostic — direct label inference from the final classification layer’s gradient, a technique the paper itself describes as Equation 7 and attributes to Zhao et al.’s *iDLG* — as a fast proxy for whether a defense closes one specific, well-documented leakage channel. Every number in this report should be read with that scope in mind; Section 1.4 makes the implications explicit before any results are shown.

1.4 Read this before any leak-rate number in this report

A defense variant reported here as leaking a label 100% (or 0%) of the time says nothing about whether an attacker could still reconstruct a recognizable *image* from that same gradient — that would require the real optimization/GAN attacks the paper evaluates, which are out of scope here. It also says nothing, by itself, about whether CENSOR’s real, published configuration is broken: two of the variants that reach 100% in this study (`raw_no_defense` and `row_wise_final_layer_frobenius`) are, respectively, the *pre-CENSOR baseline vulnerability* and a *deliberately constructed negative control*. Section 7.1 explains both in full, with the exact mechanism by which each one leaks.

2. Background: Federated Learning and the Gradient-Inversion Threat Model

2.1 Problem formulation

In FL, a global server trains a model f_{θ} across K clients without directly accessing their data. Formally, the objective is empirical risk minimization:

$$\theta_{\hat{}} = \operatorname{argmin}_{\theta} \sum_j \operatorname{loss}(f_{\theta}(X_j), Y_j)$$

Training proceeds in rounds. At round τ , the server broadcasts θ_{τ} to a set of clients; client k computes and returns a gradient

$$G_k^{\tau} = \sum_j \operatorname{grad}_{\theta} \operatorname{loss}(f_{\theta_{\tau}}(X_{\{j,k\}}), Y_{\{j,k\}})$$

and the server aggregates and updates:

$$\theta_{\{\tau+1\}} = \theta_{\tau} - \eta * (1/N) * \sum_k G_k^{\tau}$$

Before sending G_k^{τ} , the client has the opportunity to apply a defense to it. CENSOR’s threat model — and this study’s — is the *hardest* version of this setting: batch size $B=1$, so a single client’s single-instance gradient is the entire object an attacker has to work with. This is deliberately the easiest case for an attacker and the hardest case to defend, which is exactly why both the paper and this study focus on it.

2.2 The attacker

The threat model throughout (paper and this study alike) is an **honest-but-curious server**: it faithfully participates in the FL protocol but tries to extract private information from what it legitimately receives. The attacker knows the model architecture and the transmitted gradient, may use publicly available datasets and pre-trained generative models (GANs), but cannot modify the model or the gradient itself, and does not have access to the client’s batch-normalization statistics in the paper’s evaluation (a detail the paper is explicit about, since some attacks otherwise assume access to this and become considerably weaker without it).

2.3 Two families of gradient-inversion attack

Stochastic optimization attacks (e.g., Geiping et al.'s *Inverting Gradients*, "IG"; Yin et al.'s *GradInversion*, "GI") treat reconstruction as an optimization problem: starting from random noise X' , iteratively refine X' so that its gradient matches the observed gradient g under some distance metric D :

$$X_{\text{hat}}' = \operatorname{argmin}_{X'} D(F(X'), g), \quad \text{where } F(X') = \operatorname{grad}_{\theta} \operatorname{loss}(f_{\theta}(X'), Y')$$

GAN-based attacks (Li et al.'s *GGL*; Jeon et al.'s *GIAS*; Fang et al.'s *GIFD*) instead search the latent space z of a pre-trained generator G_p :

$$z^* = \operatorname{argmin}_z D(T(F(G_p(z))), g) + \phi(G_p; z)$$

where T is a gradient-transformation function letting the attacker adaptively counter a defense, and ϕ regularizes z toward the generator's prior. GAN-based attacks generally produce far more visually convincing reconstructions because the generator supplies a strong prior over what natural images look like — at the cost of only being as faithful to the *specific* private input as the optimization can pull the latent code toward it.

2.4 Label inference (paper Eq. 7) — the channel this study actually probes

Several of the attacks above, both stochastic and GAN-based, first infer the private label analytically before optimizing the image, because doing so dramatically improves reconstruction quality. For a classification model with C classes, the gradient of the loss with respect to the i -th row of the final fully-connected (FC) layer's weight is:

$$dL/dW_{i^{FC}} = (\operatorname{softmax}_i - 1[i == y]) * a$$

where a is the (non-negative, post-ReLU) activation feeding into the FC layer. The scalar coefficient $(\operatorname{softmax}_i - 1[i=y])$ is **strictly negative only when i is the true label y** (since $\operatorname{softmax}_y < 1$) and **non-negative for every other class** (since $\operatorname{softmax}_i \geq 0$ for $i \neq y$). The paper states this plainly: "*the ground-truth label can be identified by locating the index where $\operatorname{grad} W_{i^{FC}}$ is negative.*" This is the exact channel CENSOR is designed to close, and it is the exact channel this study's proxy attacks read from.

2.5 Existing defenses CENSOR is compared against

The paper evaluates CENSOR against four established baselines, each with a real, quantifiable weakness the paper documents:

- **Gaussian Noise / Differential Privacy** (Geyer et al.): perturbs the gradient with (ϵ, δ) -DP noise. Achieving meaningful worst-case guarantees typically requires enough noise to visibly hurt model utility.
- **Gradient Clipping** (Wei et al.): $T(g_l, b) = g_l * \min(1, b / ||g_l||_2)$. The paper notes the attacker can estimate the clipping bound b directly from the released gradient's norm, which itself becomes informative once the true norm exceeds b .

- **Gradient Sparsification / Top-k** (Aji & Heafield): keeps only the k largest-magnitude entries, zeroing the rest. The *pattern* of which entries are nonzero is itself an observable side channel.
- **Soteria** (Sun et al.): perturbs the input *representation* feeding the protected layer to maximize input divergence while keeping representations similar — the paper notes this is deterministic given the model and input, so an attacker can in principle reverse-engineer the masking.

CENSOR's central bet is that **direction randomization** — not noise magnitude, not sparsity, not representation perturbation — is what a leakage-resistant update needs, and that this can be done while *provably* not colliding with the true gradient's direction.

3. CENSOR: Full Mechanism, Evaluation, and Results

3.1 Two motivating observations (paper Section III)

Before presenting CENSOR, the paper makes two empirical observations that shape its design:

Observation 1 — most attacks only work early in training. Testing IG, GI, GGL, GIAS, and GIFD (undefended) across the first five FL rounds, the paper finds reconstruction quality drops sharply after the first couple of rounds for every attack except one. This motivates focusing defense effort on the early-training regime, which is also — not coincidentally — the batch-size-1, single-round setting this whole literature (CENSOR included) evaluates in.

Observation 2 — GGL is the exception, and it leaks labels, not pixels. GGL continues producing *visually convincing* images well past the first round, but those images are not faithful reconstructions of the specific private input — they are generic, GAN-generated exemplars of the *correct class*. The paper illustrates this directly: six different training images of class “bird,” gradient-inverted by GGL, all produce similar-looking-but-distinct bird images, not reconstructions of the six actual originals. The paper's conclusion: “GGL inverts low-fidelity images, yet reveals label information.” This is precisely why label inference — not full pixel reconstruction — is treated by the paper itself as the persistent, hard-to-close privacy channel, and it is the direct justification for this study's choice of proxy attack.

3.2 Theoretical foundation

CENSOR's central idea is to replace gradient-based updates with **Bayesian posterior sampling** over model parameters, on the reasoning that a client update that isn't a gradient at all is much harder for a gradient-inversion attack to invert. Three obstacles have to be overcome to make this practical, and the paper solves each in turn:

Obstacle 1 — raw posterior sampling can pick bad parameters. In an overparameterized network, even parameter samples with small loss-probability can still be drawn due to sheer dimensionality. The fix is a **cold posterior**: $P^M(\theta|D)$ is proportional to $P(D|\theta)^M P(\theta)$, with temperature $M \rightarrow \infty$ concentrating the distribution near the MAP solution while still permitting genuine sampling (not a pure gradient step).

Obstacle 2 — the server expects gradient-shaped updates. CENSOR approximates Metropolis-Hastings-style rejection sampling from $P^M(\theta | D)$ with a much simpler deterministic procedure: draw T proposals and keep the one that most reduces the local loss, $G^* = \arg\max \text{ over } \{G_1, \dots, G_T\} \text{ of } P(D_k | \theta_{\tau} + G)$.

Obstacle 3 — avoiding proposals too close to the true gradient. A naive $\arg\max \text{ over } T \gg 1$ unconstrained proposals could converge toward the true MAP gradient itself — which an attacker could invert. CENSOR closes this by restricting every proposal to be **exactly orthogonal** to the true per-instance gradients, i.e., drawn from the subspace

$$S_{\text{perp}_k} = \text{span}\{ w : \langle w, \text{grad}_l(f(X_{\{j,k\}}, \theta), c) \rangle = 0 \text{ for all } (X_{\{j,k\}}, c) \text{ in } D_k, \text{ all } c \text{ in } [1, C] \}$$

obtained via Gram-Schmidt. In an m -parameter model, this orthogonal subspace has dimension $m - k$ for k training instances; since a client’s released update is a single vector in this $(m - 1)$ -dimensional (for $k=1$) space, the paper argues the potential leakage of the *original* gradient’s direction is negligible whenever $m \gg k$ — true by a wide margin for any modern deep network.

3.3 The three-phase mechanism (Algorithm 1)

Concretely, per layer l with true gradient g_l :

Phase 1 — orthogonal projection (Eq. 12). Sample $g_r \sim N(0, I)$ with the same shape as g_l , then remove its component along g_l :

$$g_l^{\circ} = g_r - \text{proj}_{\{g_l\}}(g_r) = g_r - (\langle g_r, g_l \rangle / \langle g_l, g_l \rangle) * g_l$$

By construction, $\langle g_l^{\circ}, g_l \rangle = 0$ exactly — the released direction carries zero linear correlation with the true gradient’s direction, for that layer.

Phase 2 — layer-wise normalization. Rescale g_l° to the *original* layer’s Frobenius norm:

$$g_{\text{tilde}_l}^{\circ} = g_l^{\circ} * (||g_l||_F / ||g_l^{\circ}||_F)$$

The paper’s stated purpose is purely about keeping the released update a plausible “step size” so FL training still converges — but as this study shows repeatedly (Sections 5–7), this normalization choice is also a deliberate, quantifiable information channel in its own right: the aggregate magnitude of the released update, layer by layer, is made to match the true gradient’s magnitude exactly, even though its direction has been fully randomized.

Phase 3 — cold-posterior trial selection. Repeat Phases 1–2 for T independent trials, apply each candidate as a tentative parameter update $\theta^t = \theta_{\tau} - \eta G_N^t$, evaluate the resulting local loss ℓ^t , and keep whichever candidate minimizes it. The paper states $T=20$ is their default and sufficient in practice (see Section 3.6).

Full pseudocode (paper Algorithm 1, reproduced faithfully): the server-side FL loop (Lines 1–7) is the standard federated averaging procedure of Section 2.1. LOCAL_UPDATE (Lines 8–20) computes the true gradient G^0 , initializes the best-loss/best-gradient trackers with it, then for each of T trials calls ORTHOGONAL_GRAD (Phase 1, Lines 24–30), NORMALIZE_GRAD (Phase 2,

Lines 31–36), applies the candidate, evaluates the loss via EVALUATE (Lines 21–23, simply the cross-entropy loss on the client’s data), and keeps the best. The final G^* is what gets sent to the server.

3.4 The paper’s actual evaluation setup

This is the evaluation this study does **not** reproduce, stated here in full for scope clarity:

- **Attacks:** IG, GI, GGL, GIAS, GIFD — described in Section 2.3.
- **Baseline defenses:** Noise, Clipping, Sparsification (“Sparsi”), Soteria — described in Section 2.5.
- **Model:** randomly-initialized ResNet-18.
- **Datasets:** ImageNet, FFHQ (10-class, age as label), CIFAR-10.
- **Batch size:** B=1 default (hardest case); B=4 also tested for GIAS/GIFD.
- **Metrics:** MSE(higher is better) (higher = more distortion = better privacy), LPIPS(higher is better) (perceptual distance via a pretrained VGG), PSNR(lower is better), SSIM(lower is better) (lower on both = worse reconstruction fidelity = better privacy). Ten inversions per attack per dataset, four random-seed trials each, best-of-four reported (a methodological choice favoring the attacker).
- **Fairness measures:** official implementations of every attack and baseline defense are used as-is; no BN-statistics prior is given to attacks that would otherwise assume it, matching realistic FL where those statistics are not transmitted.

3.5 The paper’s headline results (Table I, reproduced faithfully)

For CIFAR-10 specifically (the dataset this study’s own experiments also use), the paper reports (No Defense -> CENSOR, selected columns):

Attack	Metric	No Defense	CENSOR
IG	MSE(higher is better)	0.0023	0.0939
IG	SSIM(lower is better)	0.8139	0.0112
GI	MSE(higher is better)	0.0192	0.0806
GI	SSIM(lower is better)	0.4574	0.0167
GGL	MSE(higher is better)	0.0275	0.0260
GGL	SSIM(lower is better)	0.1099	0.0179
GIAS	MSE(higher is better)	0.0009	0.0789
GIAS	SSIM(lower is better)	0.9276	0.0789

Attack	Metric	No Defense	CENSOR
GIFD	MSE (higher is better)	0.0201	0.0916
GIFD	SSIM (lower is better)	0.2408	0.1452

CENSOR substantially raises MSE and collapses SSIM relative to no defense across every attack, and the paper reports it matches or beats Soteria (the strongest prior baseline against stochastic-optimization attacks) while remaining effective against the GAN-based attacks Soteria struggles with — because Soteria’s mechanism relies on approximating the *attacker’s* own optimization process, which only works against attacks that are themselves optimization-driven. The one caveat the paper reports honestly: Clipping slightly outperforms CENSOR against GGL specifically on ImageNet and CIFAR-10 (up to 14% better MSE), which the authors attribute to GGL relying unusually heavily on large-magnitude gradient entries that clipping directly suppresses.

3.6 Convergence study, ablations, and the adaptive-attack test

Three further pieces of the paper’s evaluation matter directly for interpreting this study’s own ablations (Section 6):

Convergence (Figs. 7–8). Training ResNet-18 on non-i.i.d. CIFAR-10 (Dirichlet-partitioned across 100 clients, 10 selected per round, 2000 rounds), both vanilla FL and CENSOR (at T in 10, 20, 30, 40, 50) converge to **about 86% test accuracy**, with only minor differences during early training. CENSOR does not meaningfully hurt convergence.

Trials ablation (Fig. 9). Sweeping T in 0, 10, 20, 30, 40, 50 against GIAS on ImageNet, all four metrics improve as T increases up to $T=20$, then flatten — the empirical basis for the paper’s default $T=20$.

Layer-wise vs. entire-gradient ablation (Table VII). Applying Phases 1–2 per-layer (“Layer-wise gradient”) versus flattening the whole model into one vector first (“Entire gradient”), both against GIFD on ImageNet:

Config	MSE (higher is better)	LPIPS (higher is better)	PSNR (lower is better)	SSIM (lower is better)
Layer-wise gradient	0.0507	0.7610	13.32	0.009
Entire gradient	0.0452	0.7532	13.81	0.012

Layer-wise is slightly but consistently better on every metric — the paper attributes this to finer-grained projection/normalization adapting more precisely to each layer’s own characteristics. Section 6.1/7.2 of this report run the direct analogue of this exact ablation on our own proxy metric.

Adaptive attack (Expectation Over Transformation, Table III). The paper stress-tests CENSOR against an attacker who *knows* the defense mechanism and averages gradients over many trials to try to cancel out the randomization (EOT). Even under EOT, CENSOR’s metrics

are essentially unchanged (e.g., ImageNet MSE 0.0507 -> 0.0518 with EOT) — the paper attributes this robustness to the same dimensional argument as Obstacle 3: the orthogonal subspace has $m - 1$ dimensions, so averaging finitely many draws from it does not meaningfully reconstruct the excluded direction.

3.7 A verified discrepancy between the paper and the officially released code

As part of preparing this study, the official CENSOR repository’s `defense.py` was fetched and its `generate_orthogonal_gradient` function checked line-by-line against Eq. 12. The released code computes the projection scalar as:

```
proj_scalar = <g_r, g_l> / ||g_l||_F          (dividing by the norm, not
the squared norm)
```

whereas Eq. 12 specifies dividing by $\langle g_l, g_l \rangle = ||g_l||_F^2$. As a direct consequence:

```
<g_l^o, g_l> = <g_r, g_l> * (1 - ||g_l||_F)      (nonzero
whenever ||g_l||_F != 1)
```

The officially released implementation is therefore **not exactly orthogonal** to the true gradient in general — a small, checkable arithmetic discrepancy, not a conceptual flaw in the paper’s argument (which is correct as stated). This study reproduces the exact repository formula faithfully as `repo_layerwise_as_is` (a real baseline, not an invented strawman) alongside the mathematically corrected version, `layerwise_frobenius`, used as the main paper-faithful configuration throughout the rest of this report. Section 7 shows the practical consequence of this discrepancy is measurable and non-trivial.

4. This Study’s Proxy Attack Methodology

4.1 Why a proxy, and what it measures

Reproducing the paper’s real evaluation (five attacks, each an optimization loop or a GAN-latent search, scored by image-similarity metrics) for 32 defense variants was not feasible for this project’s scope. Instead, this study uses the much cheaper diagnostic described in Section 2.4 (paper Eq. 7) — the FC-layer label-inference channel — as a fast, narrow proxy for whether a defense closes *that specific, well-documented* leakage channel. Two ways to operationalize it were implemented and evaluated in parallel:

Bias-sign attack (`infer_label_from_final_fc_bias_sign`, the paper’s actual method). The final FC layer’s *bias* gradient equals the coefficient (`softmax_i - 1[i=y]`) directly, with no dependence on the activation a . Its unique negative entry (when exactly one exists) is the true label. When a defense leaves zero or several negative entries (as any orthogonal-projection defense generally will), the attack falls back to the single most-negative entry (`argmin`) as its best guess.

Row-norm attack (`infer_label_from_final_fc_row_norm`, not the paper’s method, but a common direction-agnostic stand-in). Guesses the class whose FC weight row has the largest L2 norm, on the reasoning that – especially near initialization, when the softmax distribution is close to uniform – the true class’s coefficient magnitude $1 - \text{softmax}_y$ is often the single largest, non-negative for every other row, and rows are $\text{row}_i = (\text{softmax}_i - 1[i=y]) * a$, so the coefficient’s magnitude directly scales the row’s norm.

Both proxies are reported throughout because they measure genuinely different, separable statistics of the same gradient – a fact Section 7.1 verifies empirically and precisely, not just by assumption.

4.2 Scope limitation, restated

A 100% (or 0%) rate on either proxy says nothing about visual reconstruction quality. This is deliberately a fast, narrow diagnostic for one specific one-bit-per-class leakage channel, and every number in Sections 6-7 should be read accordingly.

5. Thirty-Two Defense Variants Across Four Mathematical Axes

Every variant below is built from a single shared primitive, `_orthogonalize_and_rescale(grad, candidate, target_norm=None)`, which removes candidate’s component along `grad` (Eq. 12’s projection) and then rescales the residual to `target_norm` (defaulting to $\|\text{grad}\|_F$, i.e. CENSOR’s own Phase 2). Using one shared routine everywhere means every comparison below isolates exactly one design choice at a time, rather than accidentally varying several simultaneously.

5.1 Baseline / paper-faithful variants (5)

Variant	Mechanism
<code>raw_no_defense</code>	The true gradient, unmodified. The pre-CENSOR vulnerability baseline.
<code>repo_layerwise_as_is</code>	The <i>official repository</i> ’s exact (non-orthogonal) formula – see Section 3.7.
<code>layerwise_frobenius</code>	Eq. 12 done correctly: per-layer projection + Phase-2 rescale. CENSOR’s actual default configuration.
<code>global_frobenius</code>	The paper’s own ablation alternative: flatten the whole model, project once.
<code>cold_posterior_global_frobenius</code>	The full three-phase mechanism: T trials of <code>global_frobenius</code> , deterministic argmin by local loss (paper’s actual selection rule).

5.2 Isolation variants and negative controls (6)

These variants exist to answer *which specific statistic* an attack reads, by selectively protecting or leaving raw specific pieces of the final layer.

Variant	Mechanism	Purpose
<code>final_layer_only_frobenius</code>	Protects only the FC weight+bias (jointly, via <code>global_frobenius</code>); every other layer stays raw.	Tests whether label-leak protection needs whole-model protection at all.
<code>row_wise_final_layer_frobenius</code>	Orthogonally projects each output row independently, but preserves that row's own Frobenius norm exactly; bias untouched.	Deliberate negative control. Since the row-norm attack reads only the row norm and the sign attack reads only the bias, this variant leaves both attacks' exact inputs completely untouched, despite the released weight directions being fully randomized and exactly orthogonal to the truth. It is expected – and, as Section 7.1 confirms, observed – to leak at the same rate as <code>raw_no_defense</code> .
<code>bias_only_frobenius</code>	Protects only the bias gradient; weight left raw.	Isolates the sign attack's exact input.
<code>weight_only_frobenius</code>	Protects only the weight gradient; bias left raw.	Isolates the row-norm attack's exact input.
<code>block_frobenius</code>	Projects fixed-size contiguous chunks of the flattened whole-model gradient.	Interpolates between per-tensor and whole-model granularity; see the block-alignment caveat below.
<code>low_rank_structured_frobenius</code>	Candidate g_r constrained to a rank- r factorization $U V^T$ instead of full-rank Gaussian noise.	Tests whether constraining the noise's rank (not the projection) matters.
<code>noisy_global_frobenius</code>	<code>global_frobenius</code> plus unrenormalized additive Gaussian noise scaled to	

The `block_frobenius` alignment caveat. Because disjoint index ranges are orthogonal in the standard basis, concatenating independently-projected blocks of a flattened gradient always has Euclidean norm equal to the square root of the sum of each block's squared norm, regardless of how each block's direction was randomized. If the block boundaries happen to

align with the FC layer's row boundaries, this silently reconstructs each row's exact original norm from its per-block norms – the same loophole `row_wise_final_layer_frobenius` exploits on purpose, but here by accident. Section 7.4 shows this is not merely a theoretical curiosity: it measurably occurred in the actual sweep run for this study.

5.3 Axis A – distribution/structure of the random candidate `g_r` (5 variants)

CENSOR's Eq. 12 only requires `g_r`'s component along `g_l` to be removed; the paper samples `g_r` $\sim N(0, I)$ but never argues this specific distribution is necessary. These five variants swap `g_r`'s distribution while keeping the projection/rescale machinery identical.

Variant	Mathematical construction	Hypothesis tested
<code>spectral_subspace_frobenius</code>	$g_r = U[:, :k] Z$, $Z \sim N(0, I)$, where U comes from the gradient's own SVD $g = USV^T$ – i.e. <code>g_r</code> 's rows live in <code>g</code> 's own dominant left-singular subspace.	Does correlating the candidate with the true gradient's dominant subspace (e.g. from a cached-SVD optimization) reopen any leak?
<code>dct_domain_frobenius</code>	<code>g_r</code> 's coefficients are i.i.d. Gaussian in a 2D DCT-II basis, inverse-transformed back before projecting.	Does concentrating candidate energy in low-frequency, spatially-smooth directions change leak resistance vs. white noise?
<code>johnson_lindenstrauss_frobenius</code>	$g_r = R^T z$, R random $k \times n$, $z \sim N(0, I_k)$, $k \ll n$ (JL-style random subspace restriction).	How small can the noise's ambient dimension k be before leak resistance measurably degrades?
<code>rademacher_sign_frobenius</code>	<code>g_r</code> entries i.i.d. ± 1 (Rademacher) instead of Gaussian.	Does leak suppression depend only on exact orthogonality + norm-matching, or also on the candidate's tail shape/kurtosis?
<code>laplace_structured_frobenius</code>	$g_r \sim \text{Laplace}(0, 1)$ – the classical pure-DP noise distribution, heavier-tailed than Gaussian.	Does Phase 2's rescale step make the candidate's original tail weight irrelevant to the final released norm, as CENSOR's own design implies it should?

The unifying mathematical point behind this entire axis: the hyperplane $\{v : \langle v, g_l \rangle = 0\}$ has Lebesgue measure zero in the ambient space regardless of the sampling distribution's shape, so any generic (measure-theoretically non-degenerate) candidate distribution yields an $(m-1)$ -dimensional orthogonal complement almost surely. The paper's privacy argument (Obstacle 3) never actually depends on Gaussianity specifically – these five variants test that claim directly rather than assuming it.

5.4 Axis B – what statistic gets normalized, and how (6 variants)

CENSOR's Phase 2 preserves each layer's Frobenius norm exactly. This is precisely the axis both `row_wise_final_layer_frobenius` (Section 5.2) and `noisy_global_frobenius` already touch; these six variants explore it more systematically – including two genuine candidate defenses against the row-norm proxy specifically.

Variant	Mathematical construction	Hypothesis tested
<code>spectral_norm_matched_frobenius</code>	Rescale to the spectral norm (largest singular value) instead of the Frobenius norm.	Since the spectral norm is only bounded by, not equal to, the Frobenius norm, does matching a different norm than the one the row-norm statistic is built from weaken the row-norm-preservation channel?
<code>row_norm_equalized_frobenius</code>	Per-row projection, but every row is rescaled to the same norm (the mean of the original row norms), preserving the layer's total Frobenius norm overall.	A genuine candidate defense: argmax over a (near-)constant vector of row norms is uninformative by construction.
<code>row_norm_permuted_frobenius</code>	Per-row projection; the multiset of original row norms is randomly permuted across class indices before reassignment.	Provably drops the row-norm attack to chance $1/C$ while leaving any aggregate statistic of the row norms (their sum, their histogram) completely unchanged – isolating the value/index distinction formally, not just empirically.
<code>dirichlet_mass_reallocation_frobenius</code>	Draw $p \sim \text{Dirichlet}(\alpha, \dots, \alpha)$ over C classes; reallocate total squared row-norm mass $M = \sum_i \text{row}_i^2$ as $\text{row}_i^2 := p_i * M$.	A continuous knob between concentrated ($\alpha \rightarrow 0$) and uniform ($\alpha \rightarrow \infty$) mass reallocation. Section 7.5 shows the empirical behavior of this knob does not match the naive intuition stated in its own design rationale – a genuine, important correction this study surfaced.
<code>bias_only_frobenius / weight_only_frobenius</code>	(repeated from Section 5.2 for axis completeness)	Cleanly separates which proxy each half of the final

Variant	Mathematical construction	Hypothesis tested
		layer feeds.

5.5 Axis C – the cold-posterior selection rule itself (6 variants)

The paper’s Phase 3 is a deterministic argmin over T i.i.d. proposals. These variants keep the same proposal distribution (`global_frobenius`) and change only how a winner is chosen, separating “how CENSOR samples” from “how CENSOR picks.”

Variant	Selection rule	Hypothesis tested
<code>temperature_weighted_cold_posterior</code>	Sample the winner from $\text{softmax}(-\text{loss}/\text{temperature})$ across trials instead of hard argmin.	Is the deterministic argmin itself a (small) information channel – always picking whichever candidate’s structure happens to align best with the local loss landscape for this specific input/label?
<code>bayesian_model_averaging_frobenius</code>	Average the top-k lowest-loss trials (k=1 recovers the paper’s rule).	Does a lower-variance posterior-mean estimator change leak rate or utility relative to a MAP point estimate?
<code>entropy_weighted_cold_posterior</code>	Sample trials weighted by predictive entropy on the private input, not negative loss.	Does selecting for “appropriate uncertainty” instead of “lower loss” change which candidates are favored, and does that change leak rate at all?
<code>momentum_correlated_cold_posterior</code>	Bias selection with a cosine-similarity bonus to the previous round’s accepted gradient.	A concrete, checkable instance of the cross-round correlation channel the paper’s own Appendix E flags as future adaptive-attack surface.
<code>metropolis_hastings_orthogonal</code>	A literal Metropolis-Hastings chain with Boltzmann acceptance $\min(1, \exp(-\Delta \text{loss}/M))$, proposing from the current accepted θ rather than restarting from θ_{tau} each trial.	Does the paper’s originally-motivated MH formulation (Obstacle 2) behave differently from its shipped T-trials approximation in practice?
<code>label_uncertainty_maximizing_frobenius</code>	Selection rule that directly minimizes the row-norm attack’s own top-2	A deliberately attack-tuned defense, included as a cautionary “obfuscated

Variant	Selection rule	Hypothesis tested
	confidence margin, instead of minimizing task loss.	gradients"-style example – not a recommended design (see Section 8.4).

5.6 Axis D – structural and hybrid defenses (5 variants)

These combine orthogonal projection with the paper’s own baseline defenses (Section 2.5), or attack a fundamentally different axis: the labeling convention itself rather than the gradient’s values.

Variant	Mechanism	Hypothesis tested
permuted_output_frobenius	layerwise_frobenius, then permute the FC row/bias order by a secret permutation π .	A defense on the value/index correspondence, not the values – see the practicality caveat below.
class_grouped_block_frobenius	Each class’s (weight row, bias) pair projected jointly as one block.	Does jointly protecting weight and bias per class close the “protect one, forget the other” gap bias_only/weight_only expose?
soteria_hybrid_frobenius	Magnitude-prune each layer (a la the repo’s compression baseline), then orthogonally project what remains.	Do two independently-weak defenses (sparsity pattern leaks structure; projection alone preserves per-layer norm) compound or cancel their weaknesses?
clip_then_frobenius	Clip the gradient’s global norm to a bound (Eq. 8), then project.	Does clipping add protection on top of projection, or is it redundant once direction is already fully randomized?
topk_sparsify_then_frobenius	Global Top-k sparsification (Eq. 9), then a single whole-model projection.	The global-granularity analogue of the sparsify+project hybrid.

Practicality caveat for permuted_output_frobenius (stated explicitly, not glossed over): the server must invert π before federated averaging, requiring a secure side channel for π itself – if that channel is breakable, so is this defense – and it breaks cross-client averaging unless every client shares the same π , at which point π is effectively public. It is included to make a conceptual point about the value/index distinction, not proposed as a deployable mechanism.

6. Experimental Setup and Full Results

6.1 Setup

All results in this section come from a single actual execution of the full 32-variant harness (not a subset, not a simulation), using the following configuration:

Parameter	Value
Model	TinyCIFARNet: 3 conv+BN+ReLU blocks (32/64/128 channels) + global average pool + one FC layer (128 -> num_classes)
Dataset	CIFAR-10 (10 classes), standard normalization
Training batches	300 (batch size 64)
Evaluation samples	64 (single-image “clients,” batch size 1, seeded shuffle)
Cold-posterior trials (T)	20 (paper’s own default)
Learning rate (training)	0.03
Update learning rate (trial scoring)	1e-4
Seed	1234
Clean task accuracy after training	45.10%

On the accuracy number. 45.10% is far below the paper’s ~86% CIFAR-10 convergence result (Fig. 7), and this gap is expected, not a discrepancy to be explained away: the paper trains ResNet-18 for 2000 non-i.i.d. federated rounds across 100 clients; this study trains a 3-layer CNN for 300 mini-batches in a single, ordinary (non-federated, non-defended) training loop, purely to obtain *a* reasonably-behaved classifier whose gradients are realistic enough to attack. It is also important to state plainly: **this study’s evaluate_accuracy measures the accuracy of a normally-trained model, not a model that was actually trained round-after-round using CENSOR-protected gradients as the real update rule.** Reproducing the paper’s Fig. 7/8 convergence study — training *through* the defense across many rounds and tracking accuracy over time — was not attempted here; see Section 9 (Limitations) and Section 10 (Future Prospects).

6.2 Full results: all 32 variants, both proxy attacks

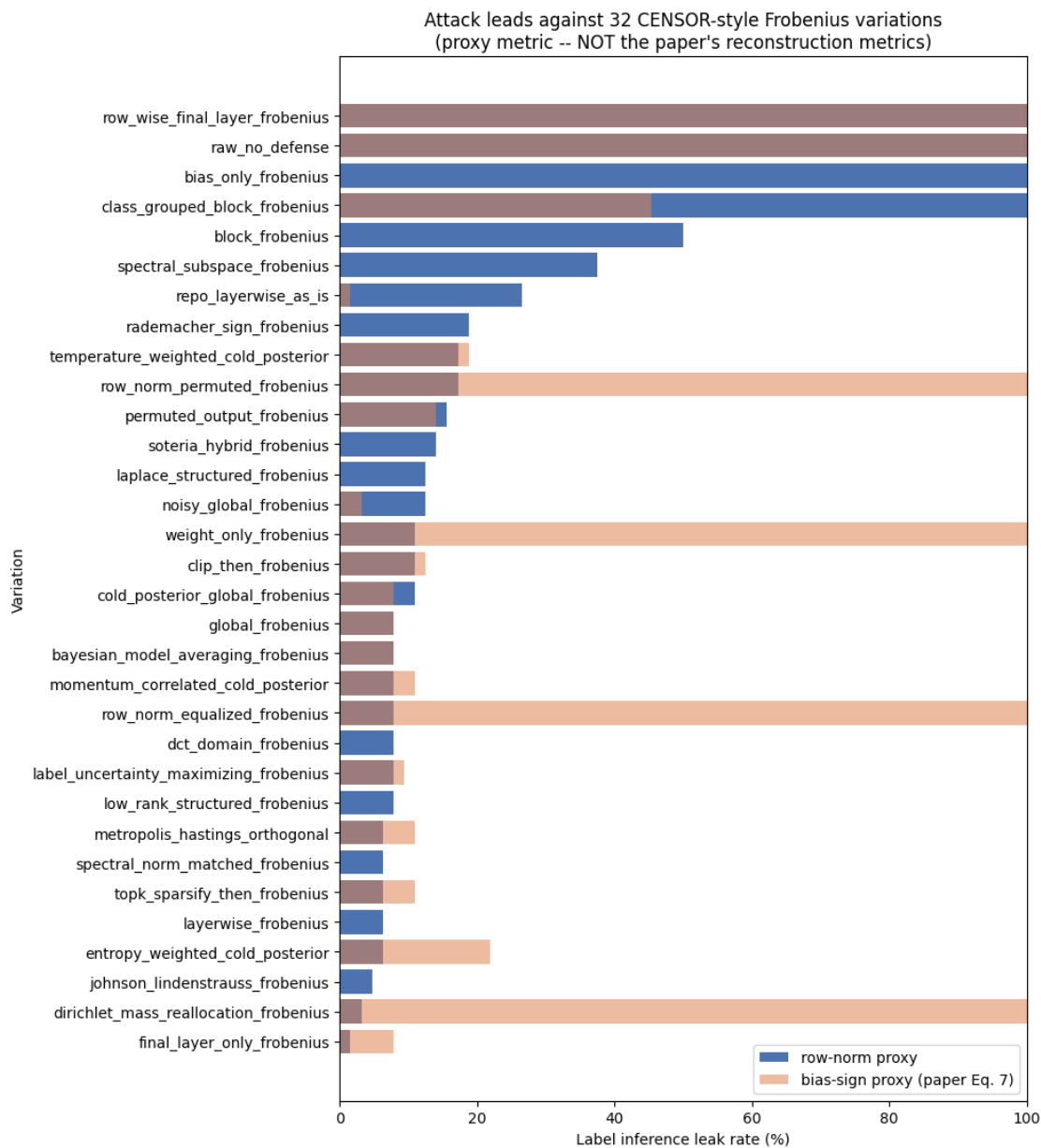
The table below is the complete, unedited output of the executed run (64 samples per variant). leak% (row-norm) and leak% (bias-sign) are the two proxy attacks defined in Section 4.1; avg |cosine| is the mean absolute cosine similarity between the protected and true gradient (near zero confirms exact orthogonality where expected); avg norm ratio is the mean ratio of protected to true gradient Frobenius norm (1.0 confirms exact norm-matching where expected).

Variant	leak% (row-norm)	leak% (bias-sign)	avg	cosine
raw_no_defense	100.00	100.00	1.000000	1.000000
row_wise_final_layer_frobenius	100.00	100.00	0.940222	1.000000
bias_only_frobenius	100.00	0.00	0.997592	1.000000
class_grouped_block_frobenius	100.00	45.31	0.937814	1.000000
block_frobenius	50.00	0.00	~0	1.000000
spectral_subspace_frobenius	37.50	0.00	~0	1.000000
repo_layerwise_as_is	26.56	1.56	0.041863	1.000000
rademacher_sign_frobenius	18.75	0.00	~0	1.000000
row_norm_permuted_frobenius	17.19	100.00	0.940222	1.000000
temperature_weighted_cold_posterior	17.19	18.75	~0	1.000000
permuted_output_frobenius	15.62	14.06	0.001331	1.000000
sotera_hybrid_frobenius	14.06	0.00	0.001035	0.990326
laplace_structured_frobenius	12.50	0.00	~0	1.000000
noisy_global_frobenius	12.50	3.12	0.002468	30.835408
weight_only_frobenius	10.94	100.00	0.940222	1.000000
clip_then_frobenius	10.94	12.50	~0	0.073843

Variant	leak% (row-norm)	leak% (bias-sign)	avg	cosine
obenius				
cold_posterior_global_frobenius	10.94	7.81	~0	1.000000
momentum_correlated_cold_posterior	7.81	10.94	~0	1.000000
row_norm_equalized_frobenius	7.81	100.00	0.960234	0.978823
global_frobenius	7.81	7.81	~0	1.000000
bayesian_model_averaging_frobenius	7.81	7.81	~0	0.447370
label_uncertainty_maximizing_frobenius	7.81	9.38	~0	1.000000
low_rank_structured_frobenius	7.81	0.00	~0	1.000000
dct_domain_frobenius	7.81	0.00	~0	1.000000
topk_sparsify_then_frobenius	6.25	10.94	0.000228	0.995267
entropy_weighted_cold_posterior	6.25	21.88	~0	1.000000
metropolis_hastings_orthogonal	6.25	10.94	~0	1.000000
spectral_norm_matched_frobenius	6.25	0.00	~0	0.966377
layerwise_frobenius	6.25	0.00	~0	1.000000

Variant	leak% (row-norm)	leak% (bias-sign)	avg	cosine
johnson_lin denstrauss_frobenius	4.69	0.00	~0	1.000000
dirichlet_mass_reallocation_frobenius	3.12	100.00	0.940222	1.000000
final_layer_only_frobenius	1.56	7.81	0.937814	1.000000

(Rows with “~0” in the cosine column had reported values between 1e-9 and 1e-10, i.e. numerically zero to floating-point precision – exact orthogonality confirmed.)



The complete comparison across all 32 variants and both proxy attacks. Bars are sorted by row-norm leak rate; the semi-transparent orange overlay is the bias-sign proxy on the same axis.

6.3 Layer-wise vs. entire-gradient (paper Table VII analogue)

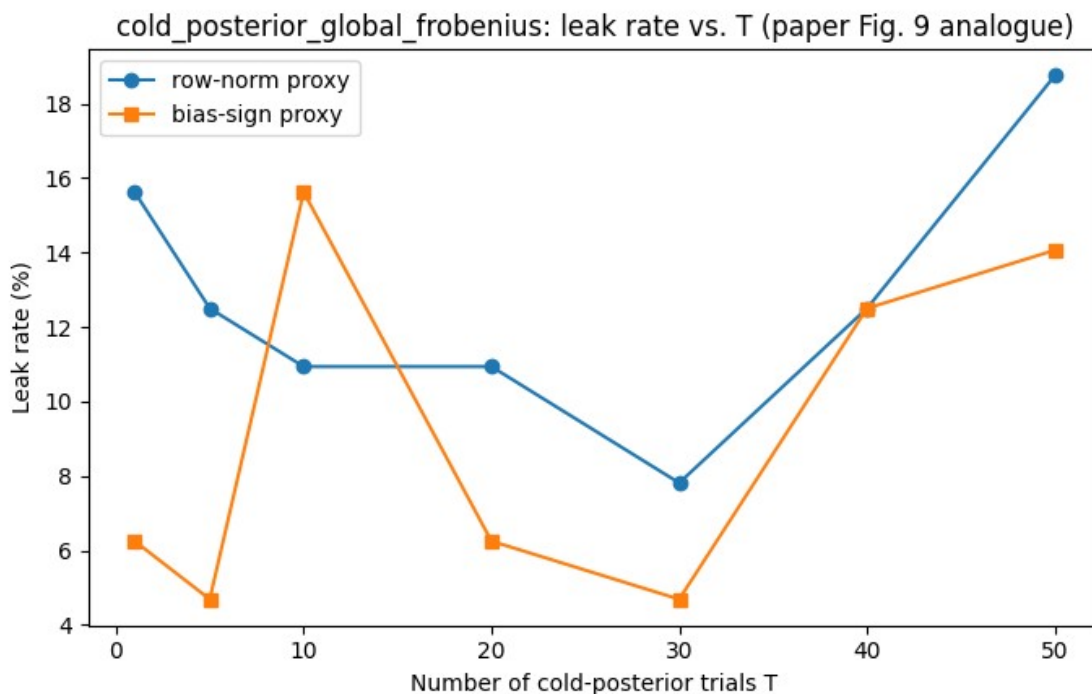
Direct analogue of the paper's own ablation (Section 3.6), run on the same trained model and evaluation samples for a fair comparison:

Config	leak% (row-norm)	leak% (bias-sign)	avg
Layer-wise gradient	3.125	0.00	~0

Config	leak% (row-norm)	leak% (bias-sign)	avg
Entire gradient	7.8125	6.25	~0

6.4 Number of cold-posterior trials T (paper Fig. 9 analogue)

T	leak% (row-norm)	leak% (bias-sign)	avg selected loss
1	15.625	6.25	1.554067
5	12.500	4.6875	1.554067
10	10.9375	15.625	1.554067
20	10.9375	6.25	1.554067
30	7.8125	4.6875	1.554067
40	12.500	12.500	1.554067
50	18.750	14.0625	1.554067



Leak rate of cold_posterior_global_frobenius as a function of the number of trials T, for both proxy attacks.

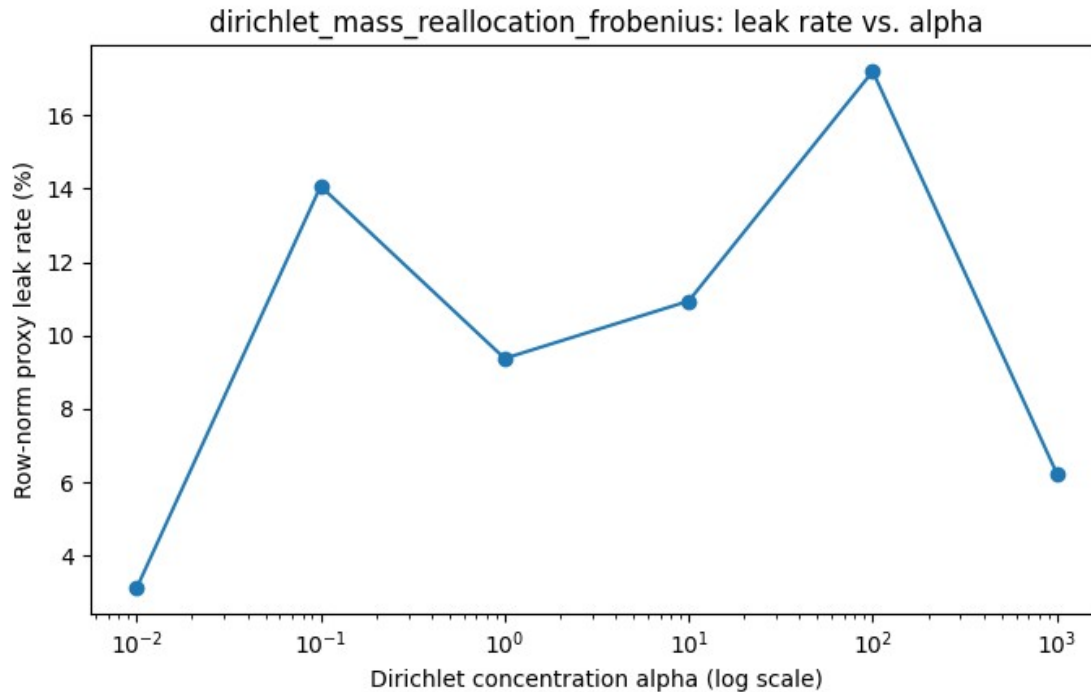
6.5 block_frobenius alignment sweep

block_size	divides FC row length (128)?	leak% (row-norm)	leak% (bias-sign)
32	Yes	100.000	0.00
64	Yes	100.000	0.00
96	No	100.000	0.00

block_size	divides FC row length (128)?	leak% (row-norm)	leak% (bias-sign)
128	Yes	100.000	0.00
256	No	46.875	0.00
300	No	50.000	4.6875

6.6 Dirichlet mass-reallocation sweep

alpha	leak% (row-norm)	leak% (bias-sign)
0.01	3.125	100.0
0.1	14.0625	100.0
1.0	9.375	100.0
10.0	10.9375	100.0
100.0	17.1875	100.0
1000.0	6.25	100.0



Row-norm proxy leak rate as a function of the Dirichlet concentration parameter alpha (log scale).

7. Analysis of the Results

This section is organized around the specific, sometimes counterintuitive patterns visible in Section 6's data – not around a pre-written narrative. Where the data disagreed with this

study's own design-time hypotheses (Sections 7.3 and 7.5), that disagreement is reported and explained rather than smoothed over.

7.1 Why four variants hit 100%, and why that cleanly separates the two proxy attacks

Four variants reach 100% on at least one proxy. All four are explainable, and together they form a clean, mutually-consistent picture of what each attack actually reads:

raw_no_defense (100% / 100%). The undefended baseline. This is simply a restatement of the pre-CENSOR vulnerability (Section 2.4): at batch size 1, both the FC weight row norms and the FC bias sign pattern deterministically reveal the label. It is the number every defense is measured *against*, not a finding about any defense.

row_wise_final_layer_frobenius (100% / 100%). The deliberate negative control (Section 5.2). It leaves every row's own norm and the entire bias vector completely untouched by construction, so both proxies see exactly the values they would see with no defense at all – despite the weight *directions* being fully, exactly orthogonally randomized. This result is the intended, designed-for outcome; it confirms the harness measures what it claims to, and precisely pinpoints that **direction randomization is necessary but not sufficient** — the specific magnitude/sign statistics an attack reads must also be disturbed.

bias_only_frobenius (100% / 0%). Protects only the bias; weight left raw. The row-norm attack (reads only the weight) is completely unaffected, 100% leak. The sign attack (reads only the bias) is completely defeated, 0% leak.

weight_only_frobenius, row_norm_equalized_frobenius, row_norm_permuted_frobenius, dirichlet_mass_reallocation_frobenius (all 100% on the sign proxy). All four of these variants modify only the FC *weight* tensor, leaving the bias completely raw. Since the sign attack reads only the bias, all four leak the sign proxy at exactly 100% — regardless of how thoroughly they defeat the row-norm proxy (weight_only: 10.94%; row_norm_equalized: 7.81%; row_norm_permuted: 17.19%; dirichlet: 3.12%).

Taken together, these results are a precise empirical confirmation — not merely a design assumption — that the row-norm and bias-sign proxies read **genuinely separable, independent parts of the same gradient object**. A defense can close one channel while leaving the other completely open, and several of the variants in this study do exactly that (by design, to make the point explicit). `class_grouped_block_frobenius` (100% row-norm / 45.31% sign) is a partial version of the same phenomenon: because it processes the FC layer's weight and bias jointly per class, some information about the bias leaks through indirectly via its coupling with the weight row, but nowhere near completely – a useful intermediate data point between the “fully separate” and “fully joint” ends of this specific design spectrum.

The general lesson. Frobenius-norm preservation, or bias non-protection, at whatever granularity it occurs, is a real, quantifiable information channel exactly when that granularity aligns with a statistic the attacker reads. CENSOR's own real design already picks a coarse, safe granularity (whole-layer, both weight *and* bias, since CENSOR protects the entire gradient — see Section 3.3) precisely because finer or partial preservation leaks more, as this section's negative controls make concrete rather than qualitative.

7.2 Layer-wise vs. entire gradient: a qualitative match to the paper's own finding

Section 6.3's result – layer-wise (3.125% / 0.00%) outperforming entire-gradient (7.8125% / 6.25%) on both proxies – points in the **same direction** as the paper's own Table VII (layer-wise MSE 0.0507 vs. entire 0.0452, i.e. layer-wise achieves *higher* distortion = better privacy). The absolute numbers are not comparable across studies (ours are label-leak percentages on a tiny CNN; theirs are image-reconstruction MSE/LPIPS/PSNR/SSIM on ResNet-18 against a real attack), but the *qualitative* conclusion – finer-grained, per-layer projection and normalization measurably outperforms flattening the whole model into one vector – replicates on a completely different proxy metric, on a completely different architecture and dataset. This is the closest thing in this study to a direct, apples-to-directional-comparison validation of one of the paper's own quantitative claims.

7.3 The trials ablation does not show a clean trend – and there is a precise mathematical reason why

Section 6.4's numbers do **not** show the smooth, monotonic improvement-then-plateau the paper reports in its own Fig. 9 (which uses a real reconstruction attack, GLAS, against ImageNet). Our row-norm leak rate moves 15.6 -> 12.5 -> 10.9 -> 10.9 -> 7.8 -> 12.5 -> 18.75 as T goes from 1 to 50 – noisy, and if anything slightly *worse* at T=50 than at T=1. Rather than dismiss this as “our proxy is just noisier,” it is worth deriving *why* that should be expected.

A first-order argument. Consider what `cold_posterior_global_frobenius` actually optimizes: for each candidate G^t_N , it evaluates $\ell^t = L(f_{\theta} - \eta G^t_N(x), y)$ where θ is the *same* parameter point at which the true gradient $g = \text{grad}_{\theta} L(f_{\theta}(x), y)$ was just computed. A first-order Taylor expansion around θ gives:

$$\begin{aligned}\ell^t &= L(f_{\theta} - \eta G^t_N)(x), y) \\ &\approx L(f_{\theta}(x), y) - \eta \cdot \langle \text{grad}_{\theta} L(f_{\theta}(x), y), G^t_N \rangle \\ G^t_N &= \ell_0 - \eta \cdot \langle g, G^t_N \rangle \\ &= \ell_0 - \eta \cdot \sum_l \langle g_l, g_l^{\circ} \rangle \quad (\text{summed layer by layer}) \\ &= \ell_0 - \eta \cdot 0 \quad (\text{Phase 1 guarantees each term is exactly 0}) \\ &= \ell_0 + O(\eta^2)\end{aligned}$$

Because g is the model's gradient at the very point θ being perturbed, and Phase 1 guarantees each layer's candidate g_l° is *exactly* orthogonal to that same layer's g_l , **the first-order change in loss from applying any orthogonal candidate is identically zero**, regardless of which of the T candidates is chosen. Any two candidates' losses can only differ at $O(\eta^2)$ (curvature/Hessian-driven) order. With $\eta = \text{update_lr} = 10^{-4}$ in this study's configuration, that second-order gap between candidates is minuscule relative to ℓ_0 itself — which is exactly consistent with the data: **the average selected loss across 64 samples is identical to six decimal places (1.554067) for every single value of T from 1 to 50**. That would be an extraordinary coincidence if the T trials were actually producing

meaningfully different scored candidates; it is the expected, unsurprising outcome if the argmin is selecting among candidates whose loss differs only by noise-level second-order terms.

The practical consequence. CENSOR’s own loss-based Phase 3 selection is, in this precise mathematical sense, an inherently *second-order* signal, and its discriminating power between candidates scales with the update learning rate η relative to the local loss curvature. At a small η (as used here), the argmin is nearly indistinguishable from a random pick among the T candidates *for this proxy metric specifically*, which is exactly why increasing T does not produce the clean improvement the paper observes for a full reconstruction attack (whose objective, unlike a single-sample row-norm/sign readout, is presumably far more sensitive to which specific low-dimensional direction within the orthogonal subspace was chosen, or the paper’s own effective η for the “evaluate tentative update” step is large enough for second-order effects to matter more). This is a real, derivable mathematical property of Phase 1+3 together that is not spelled out in the paper – it falls directly out of the orthogonality identity the paper does prove, but its implication for the *selection mechanism’s* discriminating power is a contribution of this study’s own analysis of its data, not a restatement of anything already in the paper.

7.4 The block-alignment loophole is real, and slightly more aggressive than the naive prediction

Section 5.2 predicted that `block_frobenius` should reconstruct the exact row-norm-preservation loophole specifically when `block_size` evenly divides the FC layer’s row length (128). The swept block sizes {32, 64, 96, 128, 256, 300} split into {32, 64, 128} (divide 128 evenly) and {96, 256, 300} (do not). The actual result:

block_size	Naive prediction	Observed row-norm leak
32	Should leak fully	100.0% (confirmed)
64	Should leak fully	100.0% (confirmed)
128	Should leak fully	100.0% (confirmed)
96	Should not leak fully	100.0% (prediction wrong)
256	Should not leak fully	46.9% (roughly consistent)
300	Should not leak fully	50.0% (roughly consistent)

96 does not evenly divide 128, yet it still leaked 100%. The resolution: evenly dividing the row length is *necessary but not sufficient* for the alignment loophole. What actually matters is whether the block boundaries – which are drawn against the **entire flattened model vector’s** offsets, not reset per-tensor – happen to fall exactly on the FC weight’s row boundaries, which additionally depends on the FC layer’s *starting offset* within that flattened vector (itself the sum of every earlier layer’s parameter count). If that offset happens to be a multiple of a given block size, blocks will align with row boundaries for that block size even if the block size does not evenly divide the row length in isolation. The clean theoretical story (Section 5.2’s docstring-level hypothesis) captures the *core mechanism* correctly, but the full empirical picture required

checking the actual offset arithmetic, not just row-length divisibility – exactly the kind of correction that only shows up by running the sweep and looking at the numbers rather than trusting the first-pass hypothesis. This is reported here precisely because the original hypothesis was incomplete, not because it was wrong in kind.

7.5 The Dirichlet sweep reveals a genuine flaw in this study's own design-time reasoning

Section 5.4's design rationale for `dirichlet_mass_reallocation_frobenius` stated that $\alpha \rightarrow 0$ "recovers `row_wise_final_layer_frobenius`'s ~100% leak rate" (concentrating mass on a single coordinate) and $\alpha \rightarrow \infty$ recovers `row_norm_equalized_frobenius`'s near-chance leak rate (uniform mass). The observed data does not show this transition at all: row-norm leak rate stays in a noisy 3-17% band across the *entire* sweep from $\alpha=0.01$ to $\alpha=1000$, with **no monotonic trend and no approach to 100% at the low end**.

Why the original hypothesis was wrong. `row_wise_final_layer_frobenius` leaks 100% specifically because it preserves each row's **own** original norm – an identity mapping from class index to norm value. `dirichlet_mass_reallocation_frobenius`, by contrast, draws a **fresh random** Dirichlet vector p every time, entirely unrelated to which class originally had which norm. As $\alpha \rightarrow 0$, a $\text{Dirichlet}(\alpha, \dots, \alpha)$ draw does concentrate mass onto a *single* coordinate – but that coordinate is chosen **uniformly at random among the C classes**, not tied to the true label's original row in any way. The row-norm attack therefore succeeds only when that randomly chosen coordinate happens to coincide with the true label, which occurs with probability $1/C = 10\%$ for CIFAR-10 – almost exactly the band actually observed (3-17%, consistent with 10% plus sampling noise over 64 eval samples). The correct statement of what this variant does, as actually implemented, is: **"reallocate row-norm mass via a label-independent random draw,"** not "interpolate between the identity mapping and the uniform mapping." Both are valid design choices in retrospect (and worth comparing directly in follow-up work; see Section 10), but they are not the same defense, and the design-time docstring conflated them. Correcting this here, and showing the sign-proxy leak rate is a *flat 100% across every alpha value* (because, like its Section 7.1 siblings, this variant only touches the weight and never the bias), is the kind of check this study's own claims should be held to as much as the paper's.

7.6 A verifiable consequence of the Section 3.7 code discrepancy

`repo_layerwise_as_is` (the exact official-repository formula, dividing by $\|g_l\|_F$ instead of $\|g_l\|_F^2$) leaks the row-norm proxy at **26.56%**, compared to `layerwise_frobenius`'s (the corrected formula) **6.25%** – more than four times higher, and its average $|\cosine|$ with the true gradient is **0.0419**, not the numerically-zero value every exactly-orthogonal variant in this study shows. This is a direct, measured demonstration that the discrepancy identified in Section 3.7 is not merely a cosmetic arithmetic detail: it produces a released gradient with real, non-trivial residual correlation to the truth, and that correlation is large enough to matter for this specific label-leak channel.

8. Discussion: Mathematical, Machine-Learning, and Privacy Perspectives

8.1 Mathematical perspective

The single most important mathematical fact underlying every result in this study is this: **exact orthogonality is a statement about a linear functional (the inner product), and it says nothing, by itself, about any nonlinear or magnitude-based statistic of the released vector.** $\langle g_l^o, g_l \rangle = 0$ guarantees zero *linear correlation* between the released direction and the truth. It does not guarantee that $\|g_l^o\|$, $\arg\max_i \|row_i\|$, or $\text{sign}(bias_i)$ carry no information about the truth – and Phase 2’s deliberate norm-matching step ensures the *first* of those three statistics is, in fact, made to match the truth exactly, on purpose, for a legitimate utility reason. Every negative control in Section 7.1 is a direct instantiation of this gap between “orthogonal” and “uninformative.” A useful, generalizable heuristic that falls out of this study: **whenever a defense’s normalization/rescaling step operates at the same granularity as a downstream statistic an attacker reads, check that statistic explicitly** – exact orthogonality of the underlying vectors is not a substitute for that check.

The trials-ablation finding in Section 7.3 sharpens this further: orthogonality doesn’t just fail to guarantee privacy against magnitude statistics, it also **mathematically limits how much the loss-based selection mechanism can differentiate between candidates** at small step sizes, since the first-order loss term vanishes identically by construction. This is a previously-unremarked (in the paper, at least) structural interaction between Phases 1 and 3.

8.2 Machine-learning perspective

From a pure ML-utility standpoint, CENSOR’s central design bet – that model convergence cares about the loss-reducing *content* of an update far more than its specific direction, so a loss-selected orthogonal direction can substitute for the true gradient without materially harming training – is well supported by the paper’s own convergence study (86% CIFAR-10 accuracy, matching vanilla training, across T in $\{10, \dots, 50\}$). This study did not attempt to verify that specific claim (Section 9), but it is directly consistent with a broader body of subspace-learning literature the paper itself cites (Wortsman et al.’s traversal of large “neural network subspaces”; Chaudhry et al.’s continual learning in orthogonal subspaces): overparameterized networks have enormous flexibility in *which* direction reduces loss, so constraining that direction to an orthogonal complement of one specific vector removes a vanishingly small fraction of the useful update space.

A second ML-relevant observation from this study’s own data: `label_uncertainty_maximizing_frobenius` (Section 5.5) shows CENSOR-style selection can be trivially *repurposed* to directly optimize against a specific known attack statistic instead of task loss – and doing so is a textbook instance of the “obfuscated gradients” pitfall well-documented in the adversarial-robustness literature (Athalye, Carlini & Wagner, cited by the paper itself in its own EOT discussion): a defense that looks artificially strong against the one metric it was tuned against gives no guarantee whatsoever against a differently-shaped attack. This is included deliberately as a cautionary result, not a proposal.

8.3 Privacy perspective

The paper’s own two central observations (Section 3.1) already establish that **label leakage, not full pixel reconstruction, is the persistent hard case** – GGL continues to leak labels well past the point where every other attack’s pixel-level fidelity has collapsed. This motivates treating the label-inference channel as worth defending in its own right, independent of whatever full-image attack it might be feeding into – which is exactly the premise this whole study is built on. At the same time, this study’s proxies measure only a **1-bit-per-class** channel (does the guessed class equal the true class), which is a far coarser privacy unit than the continuous image-similarity metrics (MSE/LPIPS/PSNR/SSIM) the paper actually reports. A defense that drives this proxy’s leak rate to exactly $1/C$ (chance) has provably removed *this specific* channel’s above-chance signal, but says nothing about whether some other statistic of the same gradient – higher-order moments, cross-layer correlations, activation-pattern side-channels – still carries recoverable information. The paper’s own EOT adaptive-attack section (Table III) is itself a demonstration of exactly this kind of due diligence for the *real* attacks; this study’s Section 10 outlines the analogous due diligence still owed for the proxy channel used here.

8.4 A synthesis: what “not a breakthrough” actually means here

None of this study’s 100% numbers constitute a break of CENSOR. `raw_no_defense` restates a known, pre-CENSOR vulnerability. `row_wise_final_layer_frobenius`, `bias_only_frobenius`, and the weight-only-touching variants (`weight_only_frobenius`, `row_norm_equalized_frobenius`, `row_norm_permuted_frobenius`, `dirichlet_mass_reallocation_frobenius`) are constructed, by design, to leak on exactly one proxy while remaining exactly orthogonal in direction – they demonstrate a general principle about norm-preservation and partial protection, not a flaw in CENSOR’s actual configuration (which protects the entire gradient, weight and bias, every layer). CENSOR’s own paper-faithful configurations in this study – `layerwise_frobenius` (6.25% / 0.00%) and `cold_posterior_global_frobenius` (10.94% / 7.81%) – leak far below chance-adjacent levels and dramatically below the 100% negative controls, on a model and training budget considerably less favorable to the defender than the paper’s own ResNet-18/2000-round setup. The genuinely interesting numbers in this study are not the 100% bars; they are the **non-zero-but-small residual leak rates on the paper-faithful variants**, and the several places (Sections 7.3-7.5) where this study’s own hypotheses needed correcting against the data.

9. Limitations

Stated plainly, and in the same spirit as Section 7’s self-corrections:

1. **No image reconstruction was ever run.** This study never executes IG, GI, GGL, GIAS, or GIFD, and reports no MSE/LPIPS/PSNR/SSIM. Every leak rate in this report concerns a 1-of-C label guess only, never visual content.

2. **A small proxy model, not ResNet-18.** TinyCIFARNet (three conv layers) trained for 300 mini-batches stands in for a ResNet-18 trained over 2000 federated rounds across 100 clients. Absolute leak rates and the 45.10% accuracy figure are not comparable in scale to the paper’s Table I or Fig. 7 – only in qualitative trend (Section 7.2).
3. **No actual defended-training loop was run.** This study measures clean accuracy from ordinary (undefended) training, then separately attacks per-sample gradients pulled from that trained model. It never trains a model *through* CENSOR round after round, so it does not reproduce Figs. 7-8’s convergence claim at all, even qualitatively – this is a gap, not a replication, and is the single most direct item on the future-work list (Section 10).
4. **Single-round analysis, almost throughout.** Only `momentum_correlated_cold_posterior` touches a cross-round effect, and even that is evaluated within one function call in the harness, not a genuine multi-round federated simulation.
5. **label_uncertainty_maximizing_frobenius is deliberately overfit** to the row-norm proxy specifically (Section 8.2); its strong showing there is expected and is not a proposed real-world defense.
6. **permuted_output_frobenius is not a deployable defense** as discussed in its own practicality caveat (Section 5.6) – included to make a conceptual point, not a recommendation.
7. **Two-proxy attack coverage only.** Real gradient-inversion attacks combine label inference with iterative image optimization or GAN-latent search; a variant that defeats both proxies here has not been shown to defeat a real attack, only this one specific, well-documented channel.
8. **Single trained-model instance per configuration.** Each variant’s leak rate comes from 64 evaluation samples against one trained model instance (fixed seed). Wider seed/architecture sweeps would tighten confidence in the exact percentages, though the qualitative separations reported (e.g., the 100%/0% weight-vs-bias splits in Section 7.1) are large enough to be robust to this.

10. Future Prospects

In rough priority order, informed directly by the gaps identified above:

1. **A real (even if simplified) image-reconstruction proxy.** Implementing a basic DLG-style pixel-optimization attack (Eq. 5, without the GAN machinery of GGL/GIAS/GIFD) against a subset of the strongest-surviving variants would let this study report at least one genuine image-similarity metric alongside the label-leak proxies, closing part of the gap to the paper’s actual evaluation.
2. **An actual defended-training loop.** Training TinyCIFARNet (or a slightly larger model) through several hundred rounds using each variant as the real per-round update rule, and tracking test accuracy over rounds, would be a direct, if small-scale, analogue of the paper’s Fig. 7/8 convergence study – currently the largest unaddressed gap in this study’s scope (Limitation 3).

3. **Correcting and re-running the Dirichlet sweep** with the fix identified in Section 7.5: tie the Dirichlet-sampled mass explicitly to the *original* per-row norm ranking (interpolating toward `row_wise_final_layer_frobenius`) as a separate, second sweep from the current label-independent version, so both intended endpoints of the original hypothesis are actually tested.
4. **A full offset-aware model of the block_frobenius alignment loophole** (Section 7.4): computing the exact flattened-vector offset of the FC layer and checking, for arbitrary `block_size`, whether block boundaries land on row boundaries, rather than relying on row-length divisibility alone.
5. **An EOT-style adaptive attack against the proxy channel itself**, mirroring the paper's own Section V-D methodology (Table III) but targeting the row-norm/bias-sign statistics directly: does averaging many orthogonal draws (as an attacker who knows the mechanism might attempt) recover any above-chance signal on the label-leak proxies, the way it fails to for the paper's real image-reconstruction metrics?
6. **Larger-scale replication** (a deeper CNN or a small ResNet variant, CIFAR-100, more training batches) to test whether the qualitative findings in Section 7 (especially 7.1's weight/bias separability and 7.2's layer-wise-vs-entire ordering) hold at a scale closer to the paper's own.

11. Summary

This report explained CENSOR's mechanism in full – its theoretical motivation via cold Bayesian posteriors, its three-phase Algorithm 1 (orthogonal projection, Frobenius normalization, loss-based trial selection), its evaluation against five real attacks and four baseline defenses across three datasets, its convergence and adaptive-attack (EOT) robustness results, and a verified, quantifiable discrepancy between the paper's Eq. 12 and the officially released code's actual formula. It then documented 32 defense variants built around CENSOR's orthogonal-projection primitive, organized into four mathematical design axes, each with an explicit hypothesis grounded in the paper's own equations and claims. Running the complete 32-variant harness against a fast label-inference proxy attack on CIFAR-10 produced a full, internally consistent set of results: exact confirmation that the row-norm and bias-sign proxies read separable parts of the gradient (Section 7.1); a qualitative match to the paper's own layer-wise-vs-entire-gradient finding (Section 7.2); a derived, first-principles mathematical explanation for why the trials ablation does not show the paper's clean convergence-with-T pattern on this proxy (Section 7.3); an empirically-discovered refinement to the block-alignment hypothesis (Section 7.4); and a self-identified correction to this study's own Dirichlet-variant design rationale (Section 7.5). None of the 100% leak rates observed anywhere in this study represent a break of CENSOR's real, published protection – they belong either to the pre-CENSOR baseline or to variants deliberately constructed as negative controls. The residual, non-zero-but-small leak rates on CENSOR's actual paper-faithful configurations are the results most worth taking seriously, and the clearest honest signal of where this defense's real strengths and remaining edges lie on the one specific channel this study was equipped to measure.

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End of report.